

Enforcing Second Law Invariants in Ecosystem Simulation: Thermodynamic State Vector Non-Negative Entropy Assertion

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Sprint 034 – Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Ecosystem simulation engines often struggle to maintain physical realism over extended temporal horizons, frequently allowing non-physical states characterized by negative entropy or negative entropy generation rates. In Sprint 034, we bridge macroecological modeling and non-equilibrium thermodynamics by implementing strict runtime assertions (`src/thermodynamics/state_validator`) that enforce the Second Law of Thermodynamics within the Web of Life engine. Grounded in the Clausius inequality and open-system exergy dissipation principles, our state validator intercepts monad state transitions to verify that absolute entropy ($S \geq 0$) and entropy generation rates ($\dot{S}_{\text{gen}} \geq 0$) remain within physically admissible domains. Through rigorous test verification and monad pipeline integration, this architecture prevents numerical drift into perpetual motion artifacts, ensuring that simulated biomes strictly obey thermodynamic constraints.

1 Introduction & Theoretical Background

Simulating complex adaptive ecosystems requires accounting for energy transformations, entropy production, and exergy dissipation. Within the Web of Life framework, planetary biomes are modeled as open thermodynamic systems (“Earth Pods”) receiving solar radiation flux (Q_{in}) and radiating longwave thermal energy to space (Q_{out}).

While the First Law of Thermodynamics guarantees energy conservation, the Second Law dictates the directionality of time and biochemical processes via entropy generation ($\dot{S}_{\text{gen}}$). In discrete numerical simulations, floating-point inaccuracies or unconstrained metabolic equations can inadvertently lead to violations of fundamental physical laws, such as negative absolute entropy or entropy destruction.

To eliminate these non-physical anomalies, Sprint 034 introduces the **StateValidator** module, integrating formal mathematical invariants directly into the simulation’s monad transition pipeline.

2 Thermodynamic Formulations

For any open thermodynamic system, the entropy balance over a time interval Δt is governed by the Clausius inequality:

$$\Delta S_{\text{system}} = \int \frac{\dot{Q}}{T} dt + S_{\text{gen}} \quad (1)$$

The internal entropy generation rate $\dot{S}_{\text{gen}}$ captures irreversibilities from metabolic heat dissipation, nutrient cycling, and chemical transformations:

$$\dot{S}_{\text{gen}} \equiv \frac{dS_{\text{gen}}}{dt} \geq 0 \quad (2)$$

Concurrently, absolute entropy must respect the Third Law constraint:

$$S \geq 0 \quad (3)$$

3 Architecture & Implementation

The validation logic is encapsulated in `src/thermodynamics/state_validator.ts` and interfaced via `src/thermodynamics/types.ts`. When a state transition occurs within the monad pipeline (`ThermodynamicMonad`), the validator inspects the resulting state vector Γ :

$$\Gamma = \{E_{\text{internal}}, S, \dot{S}_{\text{gen}}, T, Q_{\text{in}}, Q_{\text{out}}\} \quad (4)$$

If either invariant fails ($S < 0$ or $\dot{S}_{\text{gen}} < 0$), a `ThermodynamicViolationError` is immediately thrown, halting the simulation before propagation of non-physical states.

4 Verification & Results

We established a comprehensive test matrix (`tests/sprint_034.test.ts`) validating boundary conditions, finite numerical checks, and violation triggers:

Test Case ID	Input Vector State ($S, \dot{S}_{\text{gen}}$)	Expected Outcome	Error Thrown / Handled
TC-S34-01	$S = 150.5, \dot{S}_{\text{gen}} = 4.2$	Pass	None
TC-S34-02	$S = -0.001, \dot{S}_{\text{gen}} = 2.0$	Fail	<code>ThermodynamicViolationError</code>
TC-S34-03	$S = 50.0, \dot{S}_{\text{gen}} = -0.1$	Fail	<code>ThermodynamicViolationError</code>
TC-S34-04	$S = \text{NaN}, \dot{S}_{\text{gen}} = 1.0$	Fail	<code>ThermodynamicViolationError</code>
TC-S34-05	$S = 100.0, \dot{S}_{\text{gen}} = \text{Infinity}$	Fail	<code>ThermodynamicViolationError</code>

Table 1: State Validator Verification Matrix for Sprint 034.

5 Conclusion

Sprint 034 establishes a rigorous thermodynamic foundation for the Web of Life engine. By embedding Second Law invariants directly into the state transition monad, we guarantee that all simulated ecosystems adhere strictly to physical laws, paving the way for stable, long-term thermodynamic and ecological simulations.

For complete source code and implementation details, visit the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.